Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

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Abstract

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Keywords: economic commensalism; environmental economics; rational incentives; commons; carbon markets.

1. Introduction

The relationship between economic activity and the natural environment has long been a subject of scholarly and social inquiry, with particular attention to the tension between the productive use of natural resources and ecological sustainability. At the theoretical level, this tension reflects a deeper limitation in conventional production theory: although land, capital, and labor are commonly treated as the fundamental inputs of production, the ecological systems that sustain them are often modeled as external constraints, fixed stocks, or sources of externalities, rather than as endogenous and dynamic components of the production process (JEHLE; RENY, 2011; STIGLITZ, 1989; DASGUPTA, 2001). As a result, environmental degradation tends to appear only after production decisions have al-

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ready been made, requiring corrective mechanisms such as taxation, regulation, property-rights allocation, or collective governance arrangements (PIGOU, 2010; HARDIN, 1968; OSTROM, 2002). In practical terms, this theoretical separation is reflected in legal and institutional arrangements that frequently impose preservation duties on private agents without fully integrating ecological preservation into their economic rationale. For example, an economic agent may be required to preserve part of a biome, even when doing so reduces the amount of productive land available and, consequently, constrains the agent's production capacity, as occurs in Brazil, where legislation restricts land use to varying degrees according to the biome (BRASIL, 2012). This requirement creates an opportunity cost for agricultural production, since land that could otherwise be used productively must be preserved in the interest of protecting a collective good. The agent therefore faces a trade-off between production and preservation, often framed as a conflict between economic growth and environmental conservation.

This dichotomy can be understood through three recurring paradigms of human–environment interaction: predation, mutualism, and extractivism. Predation denotes the unsustainable and often illegal depletion of natural resources; mutualism represents a strict preservationist ideal, in which economic use is minimized in favor of ecological integrity; and extractivism occupies an intermediate position, allowing resource use within supposedly sustainable or regenerative limits (GUDYNAS et al., 2009; ACOSTA, 2013). From an efficiency perspective, however, none of these paradigms fully resolves the production–preservation trade-off. Predation disregards ecological constraints and generates negative externalities, while strict mutualism may excessively restrict productive potential (DASGUPTA, 2001; OSTROM, 1990). Extractivism, in turn, may delay ecological collapse but remains vulnerable to regulatory failure: if too restrictive, it underuses productive resources; if too permissive, it reproduces the overuse problem it was meant to avoid (CHICHILNISKY, 2017). Therefore, natural resource allocation tends to deviate from Pareto efficiency whenever private incentives are not aligned with social and ecological costs, which motivates the search for institutional arrangements capable of reconciling productive efficiency with environmental sustainability (HANLEY et al., 2007).

This article argues that such antagonism may not be inherent to economic activity itself, but rather to the way economic theory and environmental institutions have historically conceptualized nature: not as a productive factor to be strategically managed, but as an external limit to be corrected, protected, or compensated. Against this background, the article proposes a framework of economic commensalism, in which nature is endogenized within production theory and environmental preservation is examined as a potentially rational, incentive-compatible, and productivity-enhancing strategy.

–> Mostrar que a economia tradicional não ignora completamente a natureza, mas tende a tratá-la como estoque, restrição, externalidade, renda da terra ou custo ambiental;

–> Dizer que o problema do artigo é outro: a natureza raramente aparece como fator produtivo endógeno, dinâmico e estrategicamente incorporado à decisão econômica;

–> Introduzir, com cautela, a lacuna que justificará o economic commensalism. This distinction is central to the argument developed here. The paper does not treat extraction and extractivism as interchangeable terms. Instead, it uses the gap between them to identify the conditions under which resource use moves from an economically rational allocation problem to an ecologically predatory regime, as reflected in declining future productivity, ecological threshold violations, divergence between private and social payoffs, and intertemporal welfare losses.

2. The Limits of Current Environmental Governance

—> Mecanismos atuais de governança ambiental (regulação, precificação, mercados de carbono, etc.) são insuficientes para lidar com a complexidade e a dinâmica dos sistemas socioecológicos.

—> Dados e evidências mostram que, apesar dos esforços, as tendências de degradação ambiental e perda de biodiversidade continuam, indicando falhas estruturais nos modelos de governança atuais.

—> Citar esforços internacionais (Acordo de Paris, Objetivos de Desenvolvimento Sustentável) e nacionais (políticas ambientais, leis de proteção) que não têm conseguido reverter as tendências negativas.

—> Usar dados de desmatamento, perda de biodiversidade, emissões de gases de efeito estufa, Brasil e mundo, para ilustrar a insuficiência dos mecanismos atuais.

–> Concluir que é necessário repensar os fundamentos teóricos e práticos da governança ambiental, buscando abordagens que integrem melhor a natureza como parte dos sistemas econômicos e sociais, e que sejam mais adaptativas, inclusivas e eficazes.

3. Literature Review: From Externalities to Incentives

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4. Model

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5. Results

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6. Discussion

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7. Final Remarks

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